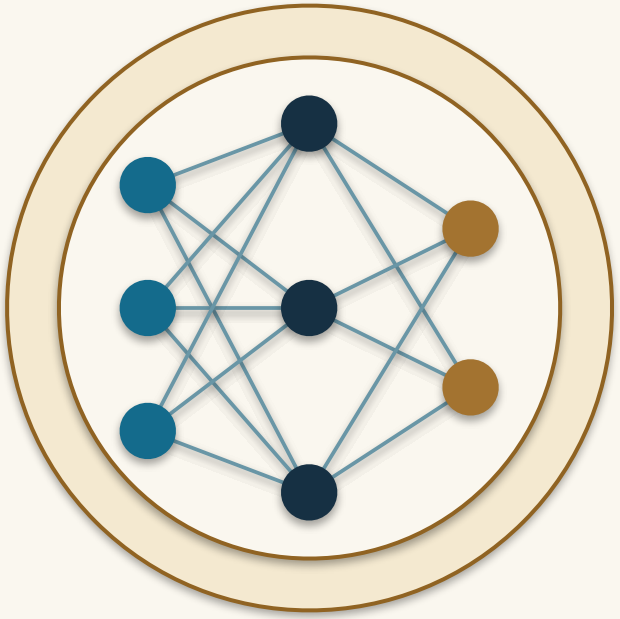


DATA • PATTERNS • INTELLIGENCE

ALEKSANDRIA

BIG DATA & CONNECTIONIST AI

From raw data to neural-network learning

A beginner-friendly introduction using handwritten-digit recognition

PROJECT PRESENTATION • STAGE 2 / V1.0

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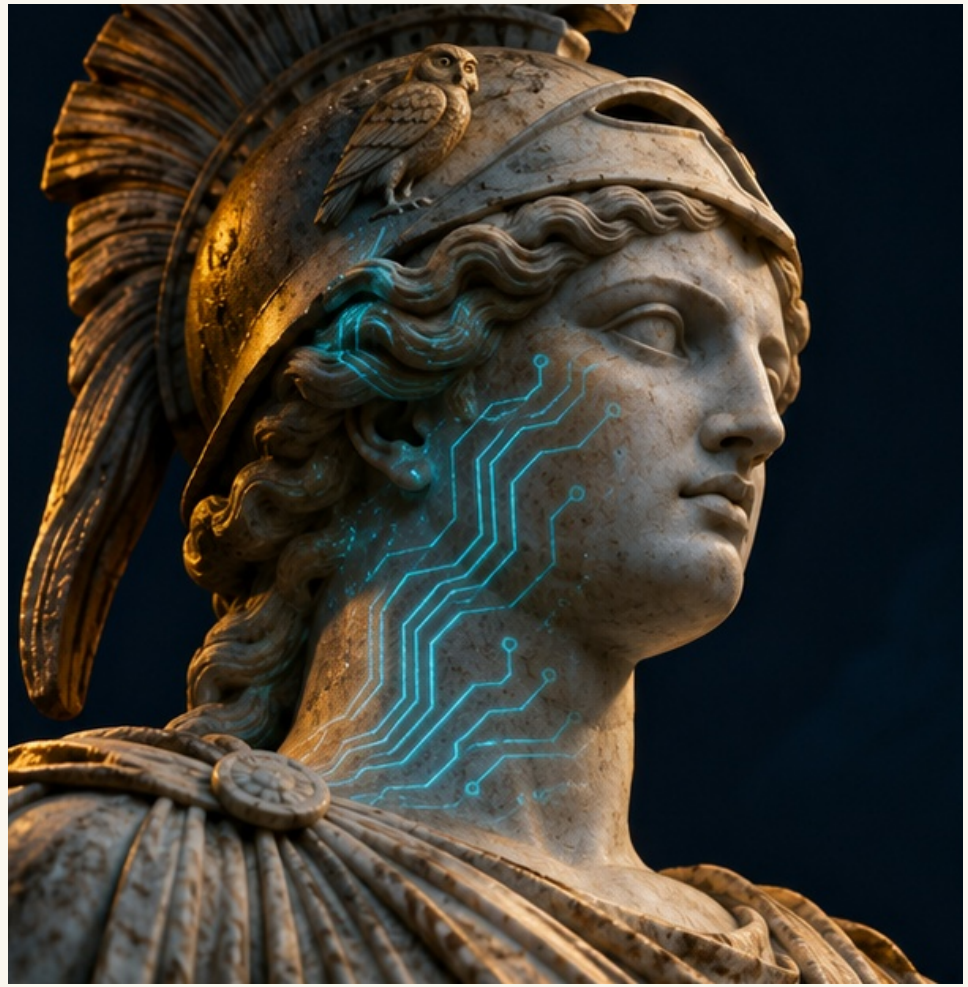
Visual design lead

How does a network recognize a digit?

The project

Learning materials for beginners

Follow a handwritten digit from a parcel scan to a network prediction. How could we tell whether the network learned?



What you will learn

Check and prepare image data → Calculate a neuron output → Explain a training update

Use unseen examples to judge how well a model recognizes handwriting.

Big Data has several challenges

Data whose scale, speed or complexity challenges conventional tools

Volume	How much data?
Velocity	How quickly does it arrive?
Variety	Which forms and formats?
Veracity	How reliable is it?
Value	What useful purpose does it serve?

Three forms of data

Form	Digit-project example
Structured	Table of image IDs and labels
Semi-structured	JSON image metadata
Unstructured	The handwritten image itself

The representation depends on the kind of data and the task.

What a parcel scanner records



A conveyor scanner captures the label image. Its log records the time and device ID. A worker can supply a checked postcode.

Link the image, the event and the checked label by a scan ID.

From a storage device to a data centre



A drive stores the scan

A server reads it when a depot requests the image.

A rack holds multiple servers

The data centre supplies their electricity, cooling and network.

Local files, network storage and the cloud

One depot

Network-attached storage (NAS) lets devices share files over the local network.

Several depots

Cloud storage lets sites retrieve scans over a wide-area network. Access depends on connectivity and permissions.

Storing and processing scans

Daily batch

Group a day of scan records by depot. Count failures and divide by the number of scans.

Live event stream

Update a five-minute failure rate as scans arrive. Alert staff while there is time to inspect the scanner.

Activity: inspect the dataset first

A Some images of 7 are labelled as 1.

B The same images appear in training and test sets.

C Almost every image comes from one writer.

What could go wrong? Suggest one fix for each problem.

Dataset fixes and their reasons

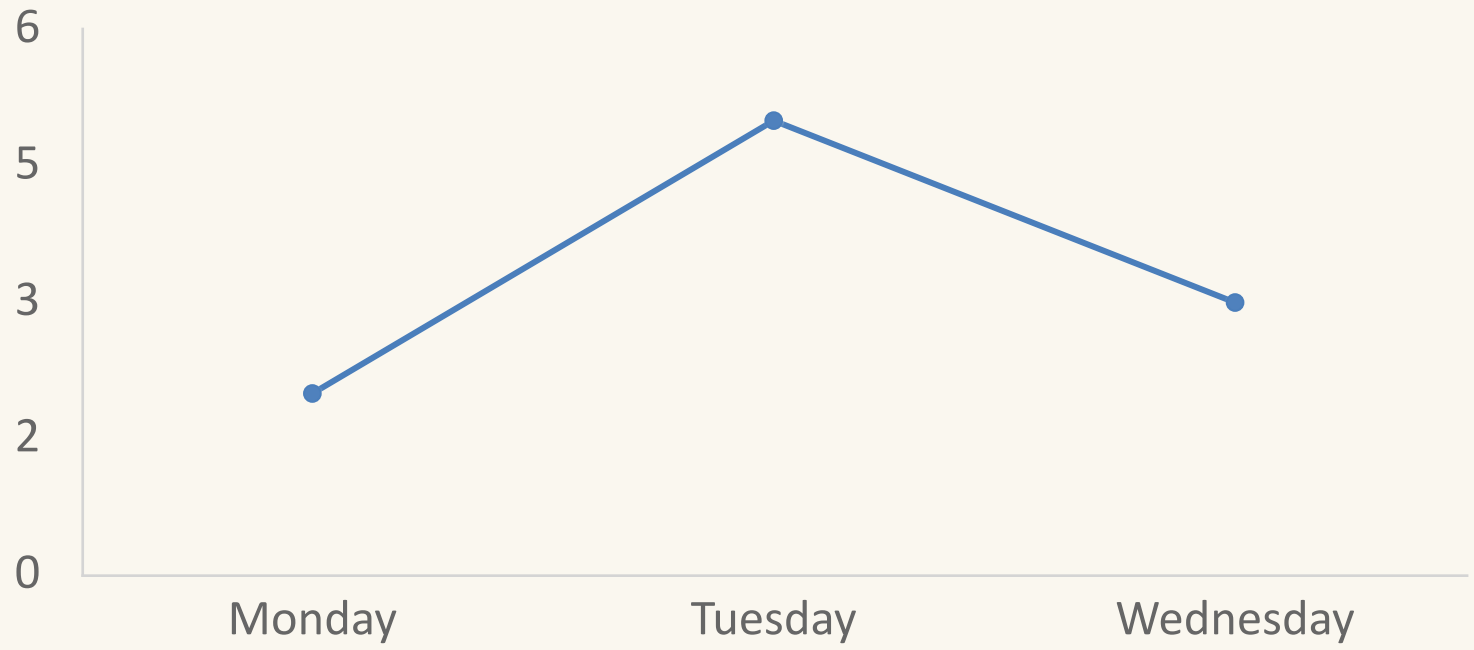
A Check labels against the original images.

B Keep duplicate images out of the test set.

C Include varied handwriting and unseen writers.

Reliable evaluation starts with how the data is prepared.

Failed scans per 100 scans



Each day covers 100 scans.

What else could explain Tuesday's peak?

Would you replace the scanner after Tuesday? What would you check first?

Two decisions at the sorting depot

Plan the next shift

Count scans and failures by hour. Compare similar shifts to decide when reviewers are needed.

Route one parcel

Inspect its cropped postcode. If a digit is uncertain, ask a person to check it.

Which task needs the original image?

Who can use a parcel scan?



Which part would you share?

For this exercise, learners need only the handwritten 5.

Prepare the classroom copy

Crop the 5 before sharing. Restrict access to the original scan.

Rules and learned connections

Symbolic rule

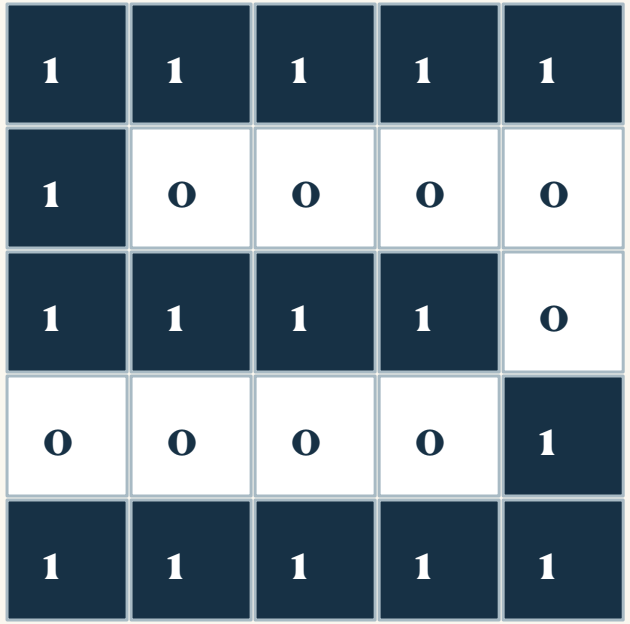
A postcode field requires five digits. A length check can flag an incomplete entry.

Connectionist model

Handwritten fives vary in shape. A network learns weighted connections from labelled images.

A sorting system can recognize a digit with a network and check the result with a rule.

An image becomes a vector of numbers



Illustrative 5 × 5 image

Read rows in a fixed order

[1, 1, 1, 1, 1,
1, 0, 0, 0, 0, ...]

25 pixel features
Target label: 5

The parts of an artificial neuron

Inputs
x_1 and x_2

Weighted sum
plus bias

Activation
then output

$$z = w_1x_1 + w_2x_2 + b$$

ReLU output = max(0, z)

A worked neuron calculation

Inputs and parameters

$x_1 = 0.8$ $w_1 = 0.5$

$x_2 = 0.2$ $w_2 = -0.25$

$b = 0.1$

Calculation

$z = 0.40 - 0.05 + 0.10$

$z = 0.45$

$\mathrm{ReLU}(0.45) = 0.45$

This is one neuron's output, not a digit label.

Activity: change only the bias

Keep the weighted inputs: 0.40 and −0.05

New bias: b = −0.50

Calculate z. Then calculate ReLU(z).

Answer: a negative sum gives zero

$$z = 0.40 - 0.05 - 0.50 = -0.15$$

$$\mathrm{ReLU}(-0.15) = 0$$

Bias shifts the sum; activation determines the output.

Which architecture fits the input?

Convolutional neural network

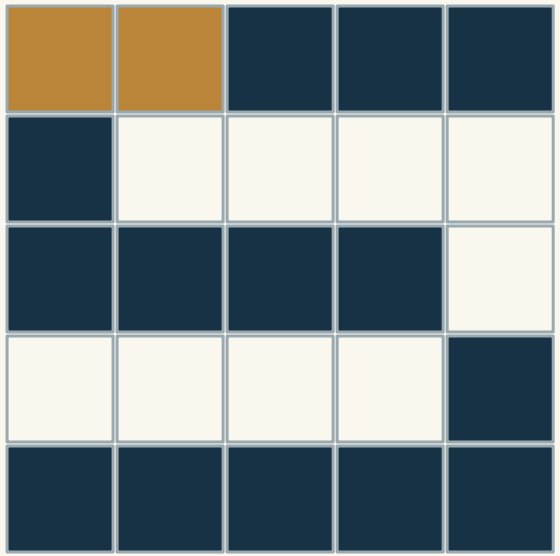
A convolutional filter looks for a stroke at different positions in an image.

Recurrent neural network

A recurrent network carries information from one step to the next, such as the order of letters in a word.

Reading a handwritten word requires both stroke shapes and their order.

Which pixel pair signals an edge?



Same digit 5

Predict the filter response

A filter with weights [1, −1] gives:

A Two ink pixels [1, 1]

B Ink beside paper [1, 0]

Calculate first pixel minus second pixel. Which pair gives the larger response?

The edge produces the larger response



A Across ink

$$1 \times 1 + (-1) \times 1 = 0$$

B Ink to paper

$$1 \times 1 + (-1) \times 0 = 1$$

This filter detects a local change. Several digits share the same edge.

The MNIST digit collection

Standard training set

60,000 labelled images. Reserve a portion for validation when comparing models.

Standard test set

10,000 labelled images for evaluating the selected model.

Each grayscale image measures 28 × 28 pixels and has one digit label.

Preparing the tutorial's digit 5

28 × 28 pixels → Divide values by 255 → Read rows in order

The model receives 784 values. The checked label 5 stays separate.

How features contribute to a digit score

A small example: two hidden units respond with 1 and 0.5.

Output for digit 5

$2 \times 1 + 1 \times 0.5 = 2.5$

Output for digit 3

$1 \times 1 + 2 \times 0.5 = 2.0$

Each class weights the same features differently. These are raw scores.

Prediction and target label are different

Hypothetical model output

Digit 3: 60%
Digit 5: 30%
Other digits together: 10%

Checked reference

Target label: 5

Predicted label: 3
This prediction is incorrect.

The prediction comes from the model. The target comes with the data.

A loss value for the tutorial prediction

The correct digit is 5. The model assigned it 30%.

$$\text{Loss} = -\ln(0.30) \approx 1.20$$

Assigning a higher probability to the correct digit would reduce this loss.

Squared error for a numerical prediction

Target 1.0, output 0.6

$$(0.6 - 1.0)^2 = 0.16$$

Target 1.0, output 0.9

$$(0.9 - 1.0)^2 = 0.01$$

Moving the numerical output closer to the target reduces squared error.

Training and inference

Training

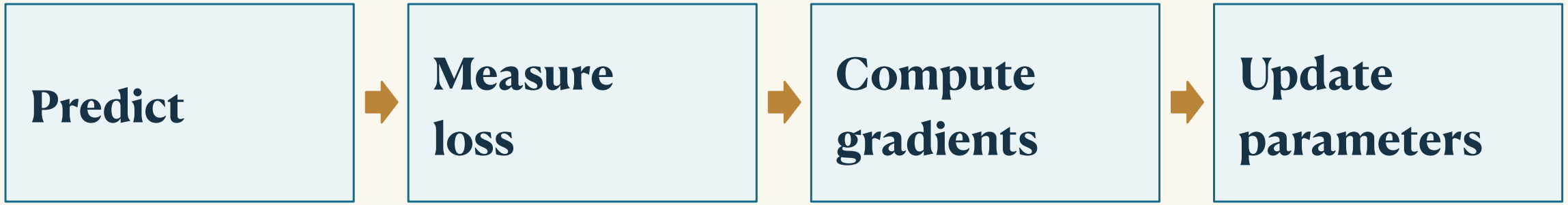
Input plus known target
Measure loss
Update weights and biases

Inference

New prepared input
Use fixed model parameters
Produce a prediction

A prediction can be made before its true label is known.

The supervised training cycle



Repeat with labelled training examples

Learning changes weights and biases.

One small parameter update

new weight = old weight − learning rate × gradient

Weight 0.40 **Learning rate 0.10** **Gradient +0.20**

Vote: will the weight rise or fall? Then calculate its new value.

Training and evaluation use different data

Training
Adjust parameters

Validation
Choose settings

Test
Evaluate the chosen model

Weight check: 0.40 − 0.10 × 0.20 = 0.38. Test the effect on unseen data.

Recognizing an image and generating one

Recognition

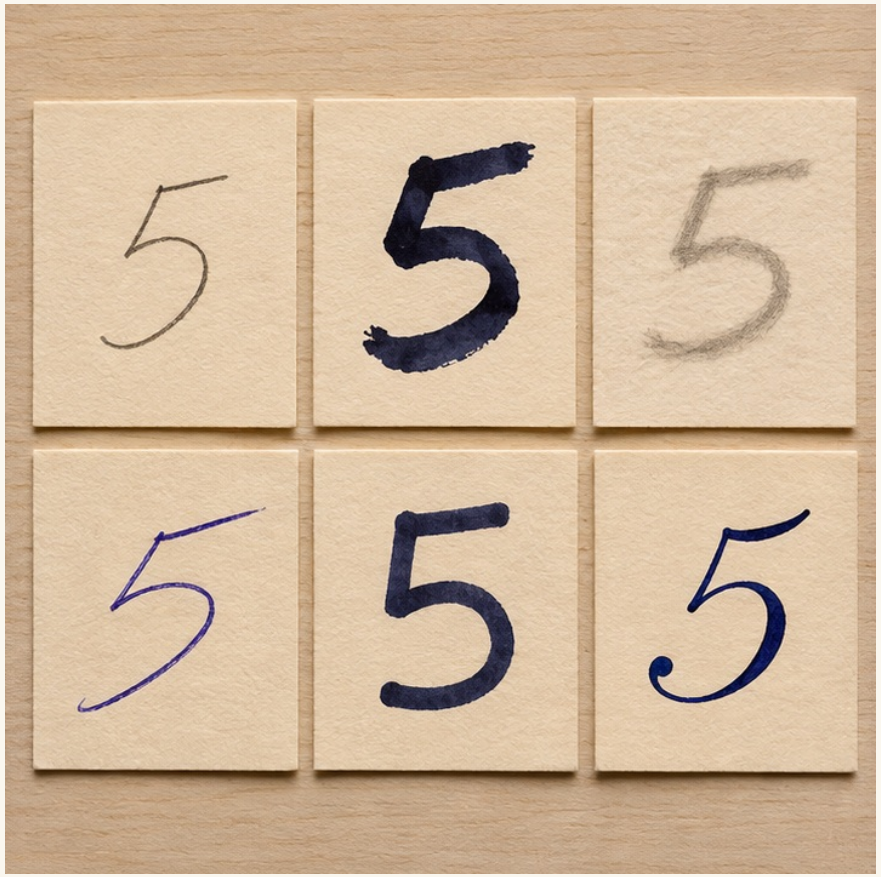
A classifier scores possible digit labels for an existing image.

Generation

A diffusion model starts with noise and progressively refines an image using learned patterns.

A generated reconstruction of a damaged manuscript can contain invented details.

Responsible use of image data



Would all six be equally easy to read?

Compare errors for faint pencil, thick ink and unfamiliar handwriting.

When the result is uncertain

Send the scan for human review. Keep personal details out of the test set.

Where do the errors fall?

	Predicted 3	Predicted 5
True digit 3	8	2
True digit 5	4	6

Which mistake occurs more often? How many of the 20 digits are correct?

What the diagonal tells us

	Predicted 3	Predicted 5
True digit 3	8	2
True digit 5	4	6

4 fives read as 3

2 threes read as 5

Accuracy = (8 + 6) ÷ 20 = 70%. Investigate the four missed fives.

Final case: a parcel goes to the wrong route

A new camera scans a handwritten 5. The sorter reads it as 3.

What should the depot do with this parcel?

What would you inspect before changing the model?

What the depot found

The label

A reviewer confirms the 5 and routes the parcel correctly.

The scan

The new crop clipped the top stroke. With the whole digit visible, the same model predicts 5.

The decision

Fix the crop. Check fresh, labelled scans before using the change across the depot.

What it took to read this 5



The camera captured the strokes. The network used patterns learned from labelled examples to score the image.

The reader needed the whole digit. Before trusting it with the next parcel, test it on handwriting it has never seen.

Thank you for your attention

Big Data and the five Vs — IBM, *What Is Big Data?*
Data quality and preparation — Google Machine Learning Crash Course, *Dataset Characteristics* and *Normalization*
Neural networks and training — Google Machine Learning Crash Course, *Neural Networks* and *Gradient Descent*
Handwritten-digit recognition — Keras, *MNIST Digits Classification Dataset*; TensorFlow, *Quickstart for Beginners*
Project examples and visuals — Team Aleksandria.